

Molecular structure inference from molecular spectra

Phase I – Pretraining / pilot

- ✓ Train on SMILES inputs, and:
 - ✓ bias a hidden state towards ChemBERTa-v2,
 - ✓ hijack a hidden state to predict functional groups,
 - ✓ augment with non-canonical SMILES.
- ✓ Pretrain on the one-atom-one-peak “quasi-NMR”

Phase II – NMR to SMILES inference training

- ✓ More careful data split (correlations due to USPTO origin)
- ✓ Train on Alberts et al., 2024 [1]
- ✓ Compare vs Alberts et al., 2025 [2]
- ✓ Compare H+C+HSQC vs H+HSQC

Phase I – Pretraining

O=C([O-])[O-].[Ca+2][:][_][o][_][<=>]C([O-])([O-])=O.[Ca+2] ■

A given SMILES structure

Phase I – Pretraining

O=C([O-])[O-].[Ca+2][:][_][o][_][⇌]C([O-])([O-])=O.[Ca+2] ■

Unsupervised trigger token (molecular embedding)

Phase I – Pretraining

$\text{O}=\text{C}([\text{O}^-])[\text{O}^-].[\text{Ca}+2][:] \text{[red box]} [\circ][\text{—}][\rightleftharpoons] \text{C}([\text{O}^-])([\text{O}^-])=\text{O}.[\text{Ca}+2] \blacksquare$

Supervised hidden state (molecular embedding)

Phase I – Pretraining

O=C([O-])[O-].[Ca+2][:][_][o][_][<=>]C([O-])([O-])=O.[Ca+2] ■

Unsupervised trigger token (functional groups)

Phase I – Pretraining

$$\text{O}=\text{C}([\text{O}^-])[\text{O}^-].[\text{Ca}+2][:][][\circ][\text{---}][\rightleftharpoons]\text{C}([\text{O}^-])([\text{O}^-])=\text{O}.[\text{Ca}+2] \blacksquare$$

Supervised hidden state (functional groups)

Phase I – Pretraining

O=C([O-])[O-].[Ca+2][:][_][o][_][<=>]C([O-])([O-])=O.[Ca+2] ■

Unsupervised trigger token (alt SMILES)

Phase I – Pretraining

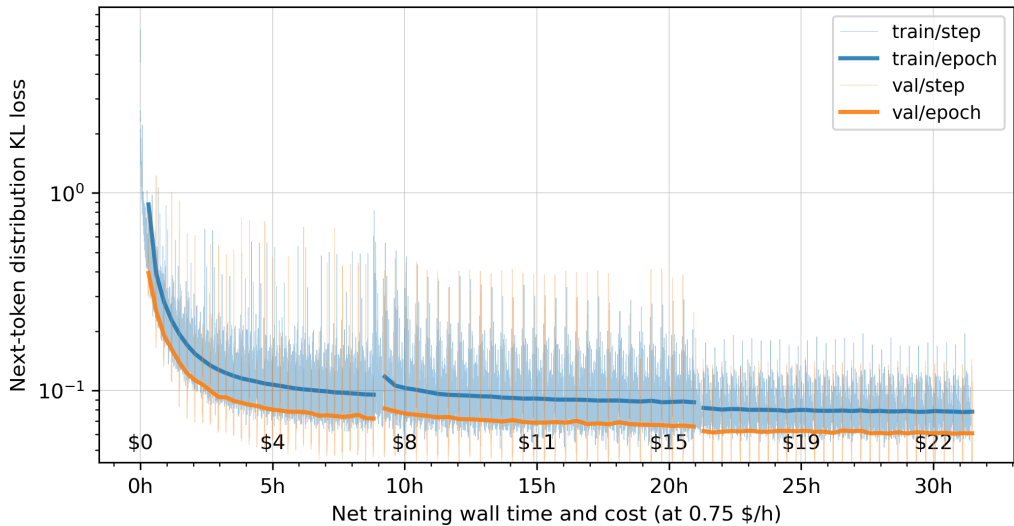
O=C([O-])[O-].[Ca+2][:][_][o][_][<=>C([O-])([O-])=O.[Ca+2] █

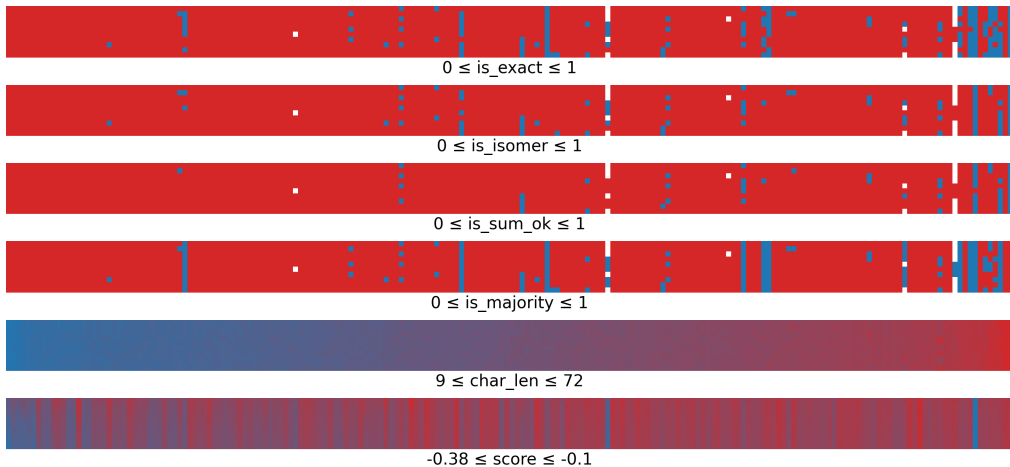
Distribution-supervised alternative SMILES

Phase I – Pretraining

$\text{O}=\text{C}([\text{O}^-])[\text{O}^-].[\text{Ca}+2][:][][\circ][][\rightleftharpoons]\text{C}([\text{O}^-])([\text{O}^-])=\text{O}.[\text{Ca}+2]$ 

Supervised “end of sequence”





low $\rightarrow$ high by: char_len

Phase II – NMR to SMILES with next-token distribution

F|C13H10N2O6

Q|2.0:5.61-5.72@89.9|1.0:8.12-8.23@126|1.0:7.02-7.13@115

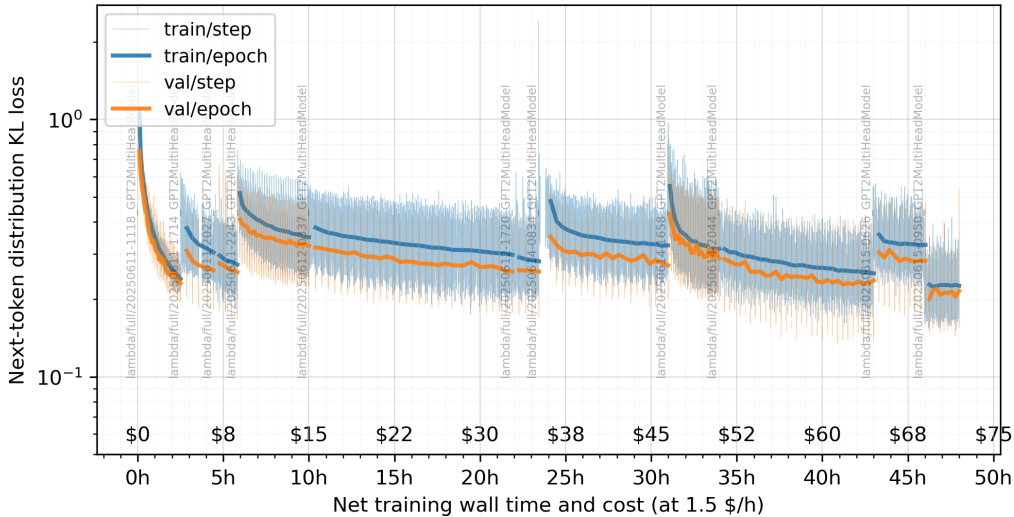
C|10@163|10@142|34@126|34@115|12@90

H|4:8.15-8.2:m|4:7.06-7.1:m|2:5.66-5.68:s

[:]_[]_[]O=[N+]([O-])c1ccc(OCOc2ccc([N+](=O)[O-])cc2)cc1

Phase II – NMR to SMILES with next-token distribution

- Training **resumed** from the model pretrained in Phase I
- 3 runs of quasi-NMR on up to ~ 600 k PubChem SMILES
- 3 runs of MestReNova NMR on up to ~ 600 k “Alberts” SMILES
- 5 runs ... with adamw, varying learning rate, no auxiliary tasks





$0 \leq \text{is_exact} \leq 1$



$0 \leq \text{is_isomer} \leq 1$



$0 \leq \text{is_sum_ok} \leq 1$



$0 \leq \text{is_majority} \leq 1$



$9 \leq \text{char_len} \leq 76$



$-0.44 \leq \text{score} \leq -0.12$



$1 \leq \text{shell} \leq 10$

low $\rightarrow$ high by: char_len

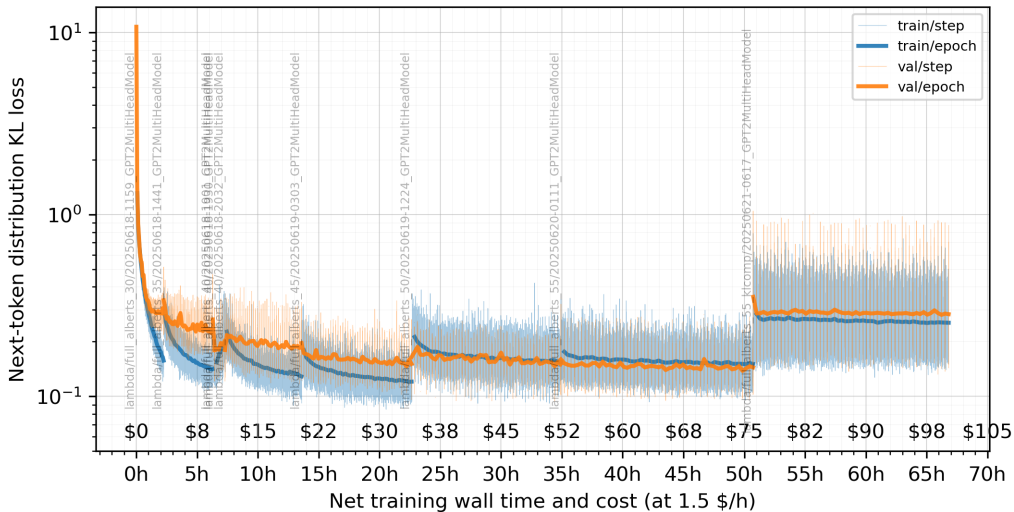
Phase I showed feasibility of next-token distribution training.

But NMR-to-SMILES stagnated.

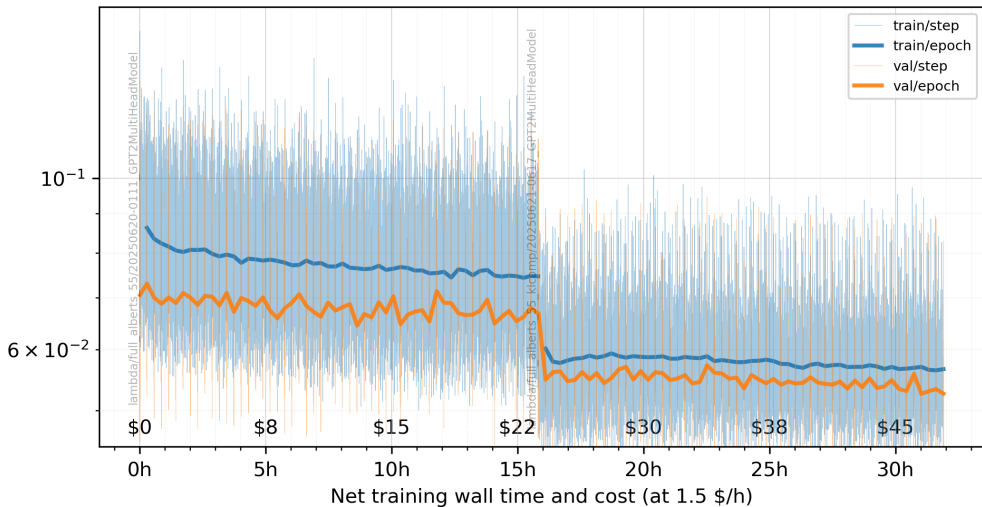
Now: Train NMR-to-SMILES from DistilGPT-2 directly.

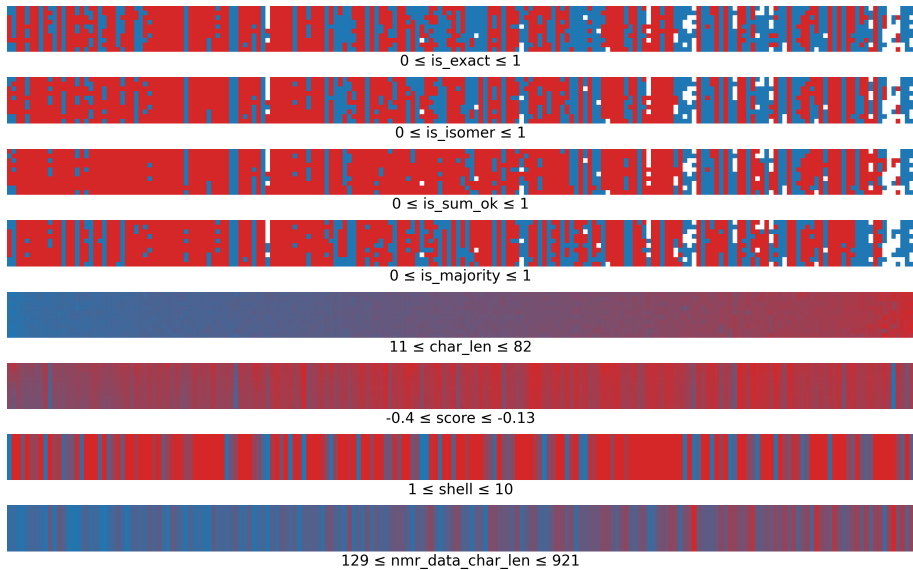
Phase II – NMR to SMILES with next-token distribution

- ~~Training resumed from the model pretrained in Phase I~~
- 7 runs with increasing maximal SMILES length (shown last time)
- additional run: KL geometric penalty with increasing length (new)

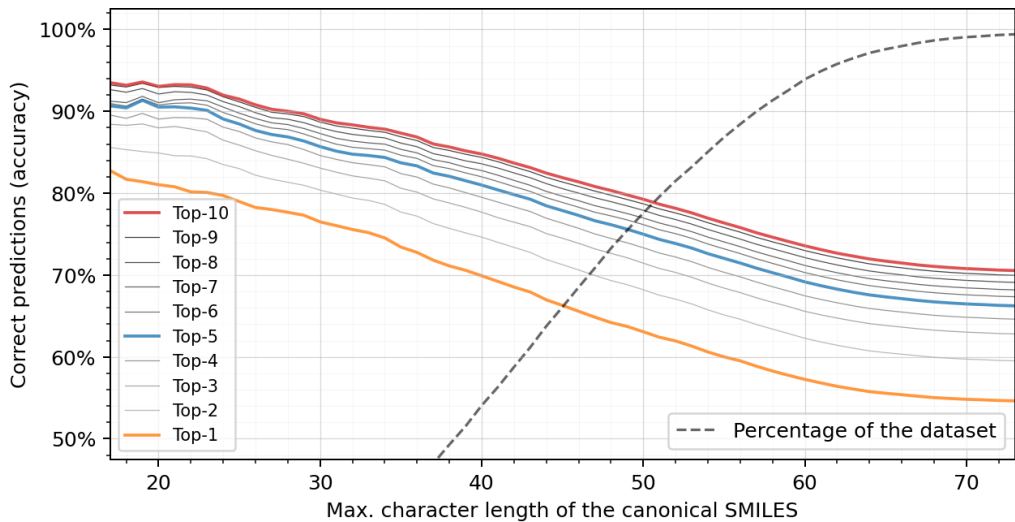


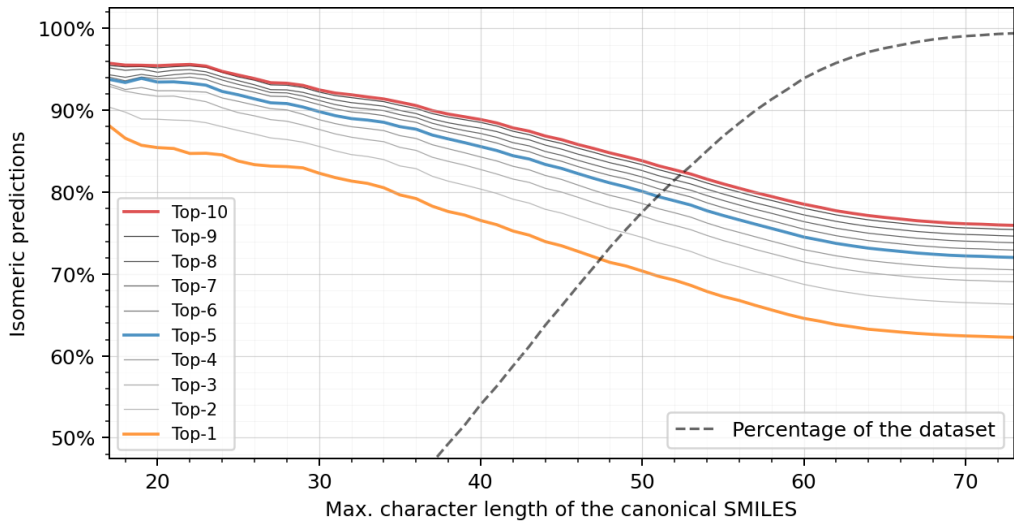
Probability mass outside of KL support





low $\rightarrow$ high by: char_len

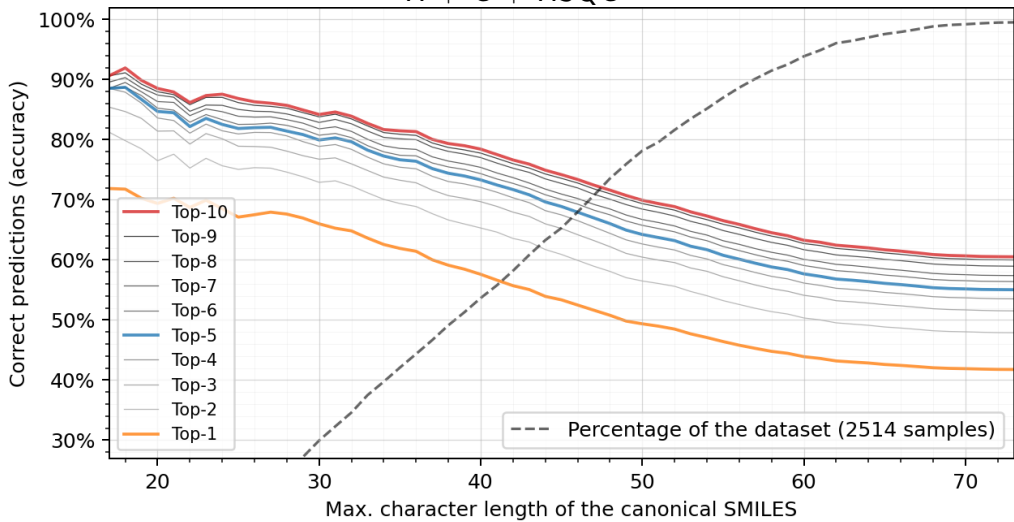




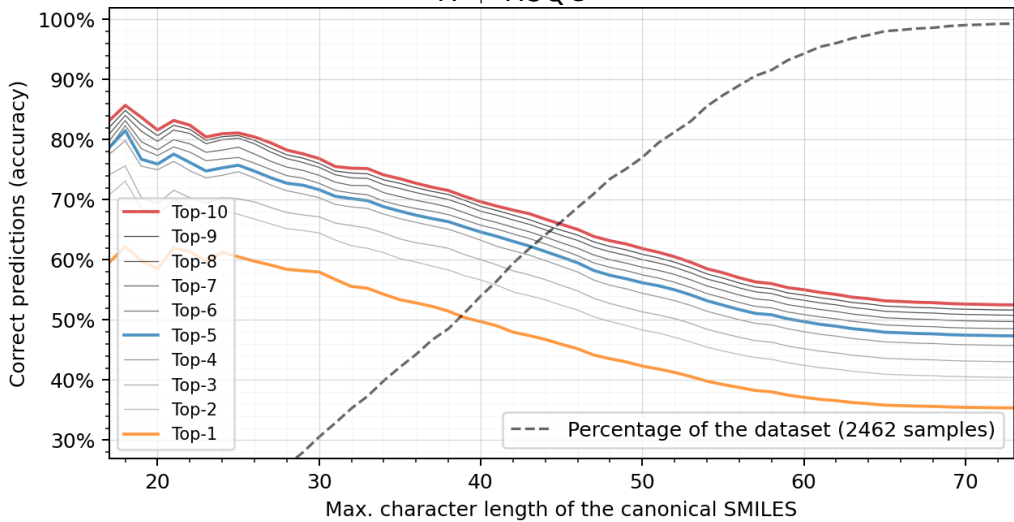
Variable-modality training and inference

- Resume training
- Switch on/off the HSQC and ^{13}C data randomly
- Fine-tune on the previously held-out “validation” subset (10%)
- Still use the test set for evaluation (10%)

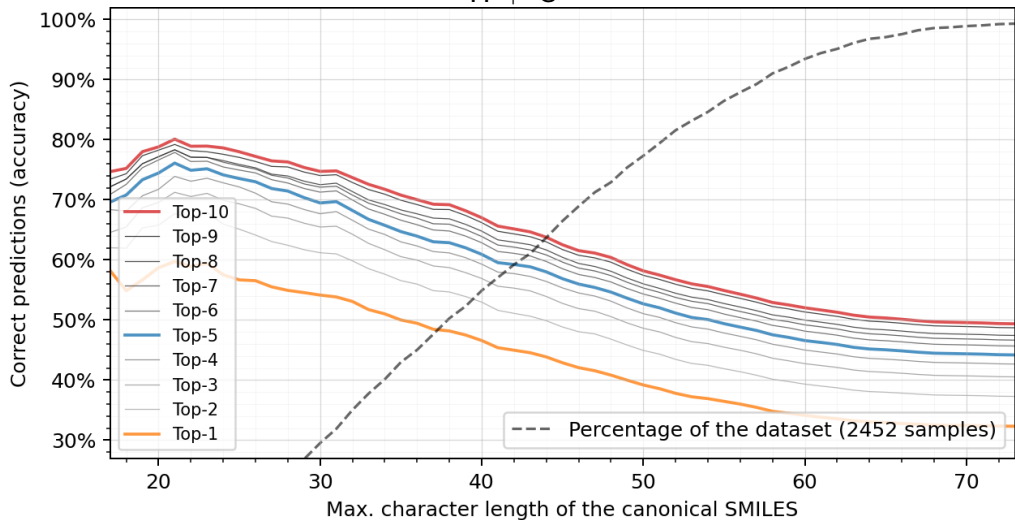
H + C + HSQC

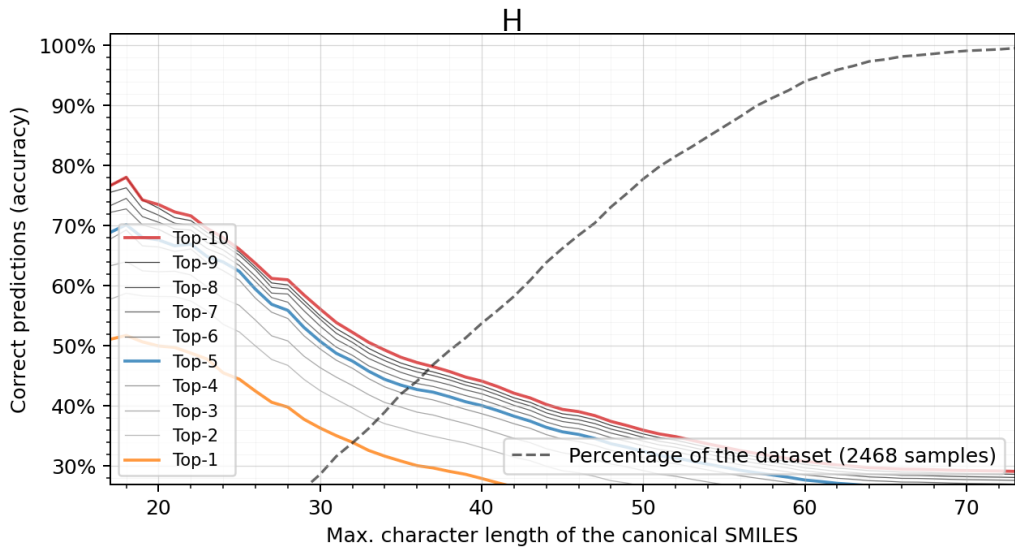


H + HSQC



H + C





Modality	Formula	Format	Tokens	Top-1%	Top-5%	Top-10%
^1H NMR	$\times$	Center		38.29%	54.67%	58.43%
	$\checkmark$	Center		53.34%	71.71%	75.09%
	$\checkmark$	Adaptive		53.39%	71.84%	75.23%
	$\checkmark$	Range		55.32%	73.59%	76.74%
^1H NMR (Aug.)	$\checkmark$	Range		51.58%	70.52%	73.94%
^1H NMR (Ens.)	$\checkmark$	Range		57.99%	76.65%	80.04%
^{13}C NMR	$\times$			37.21%	53.98%	57.45%
	$\checkmark$			51.37%	70.74%	74.32%
^{13}C NMR (Aug.)	$\checkmark$			49.02%	69.05%	72.90%
^{13}C NMR (Ens.)	$\checkmark$			53.91%	73.45%	77.72%
$^1\text{H}+^{13}\text{C}$ NMR	$\times$	Range	Own	56.88%	73.91%	76.89%
	$\checkmark$	Range	Own	64.78%	81.74%	84.43%
	$\checkmark$	Range	Shared	65.05%	82.07%	84.70%
$^1\text{H}+^{13}\text{C}$ NMR (Aug.)	$\checkmark$	Range	Shared	62.35%	80.15%	82.93%
$^1\text{H}+^{13}\text{C}$ NMR (Ens.)	$\checkmark$	Range	Shared	66.99%	84.09%	86.59%

Alberts et al. [3], Table 1

thx.

References I

- [1] Marvin Alberts, Oliver Schilter, Federico Zipoli, Nina Hartrampf, and Teodoro Laino. “Unraveling Molecular Structure: A Multimodal Spectroscopic Dataset for Chemistry”. In: *NeurIPS 2024 Datasets and Benchmarks Track*. 2024. [url].
- [2] Marvin Alberts, Nina Hartrampf, and Teodoro Laino. *Automated structure elucidation at human-level accuracy via a multimodal multitask language model*. 2025. [url].
- [3] Marvin Alberts, Federico Zipoli, and Alain C. Vaucher. *Learning the Language of NMR: Structure Elucidation from NMR spectra using Transformer Models*. 2023. [url].

References II

- [4] Alex Kendall, Yarin Gal, and Roberto Cipolla. “Multi-Task Learning Using Uncertainty to Weigh Losses for Scene Geometry and Semantics”. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* (2018). [url].

Appendix

GPT-2 reference implementation¹

Input: $x \in \mathbb{N}^{B \times T}$ (token indices)

Token embedding: $\mathbf{E}_{\text{tok}} = \text{Embedding}(x) \in \mathbb{R}^{B \times T \times d}$

Position embedding: $\mathbf{E}_{\text{pos}} = \text{Embedding}[1 : T]$

Combined: $\mathbf{H}^{(0)} = \text{Dropout}(\mathbf{E}_{\text{tok}} + \mathbf{E}_{\text{pos}})$

For each block $\ell = 1 \dots L$:

$$\mathbf{X} = \mathbf{H}^{(\ell-1)}$$

$$\mathbf{X} \leftarrow \mathbf{X} + \text{CausalSelfAttn}(\text{LayerNorm}(\mathbf{X}))$$

$$\mathbf{X} \leftarrow \mathbf{X} + \text{MLP}(\text{LayerNorm}(\mathbf{X}))$$

$$\mathbf{H}^{(\ell)} = \mathbf{X}$$

Final: $\mathbf{H}_{\text{final}} = \text{LayerNorm}(\mathbf{H}^{(L)})$

Logits: $\hat{y} = \mathbf{H}_{\text{final}} \cdot \mathbf{E}_{\text{tok}}^{\top}$ (weight tying)

¹Based on <https://github.com/karpathy/nanoGPT/blob/master/model.py>

Phase I – Pretraining: losses definitions

- $\frac{1}{n}(\frac{1}{2}\|T\mathbf{e}\|_2^2 - \log \det T + \frac{n}{2} \log 2\pi)$ – embedding ($n = 384$)
- $\frac{1}{n}(\|T\mathbf{e}\|_1 - \log \det T + n \log 2)$ – functional groups ($n = 289$)
- KL divergence, position-averaged – next-token distribution
- Causal LM = $-\log p(\text{eos} \mid \text{prefix})$ – end-of-sequence token

combined à la Kendall et al. [4]:

$$\text{minimize} \quad \sum_t \left(\frac{1}{\sigma_t^2} \text{softplus}(\text{loss}_t) + \log \sigma_t^2 \right)$$

with a trainable σ_t for each task t .

Intuition for the KL loss

Suppose

$$P \sim \text{uniform over a set} \quad \text{and} \quad Q \sim (1 - q)P + qP^c.$$

Then

$$\text{KL}(P \mid Q) = -\ln(1 - q) \approx q$$

